

## Abstract

Residential foundations are still built the way they were a century ago: crews set and strip wooden or aluminum formwork, pour concrete, wait, then repair and clean — a labor-intensive, weather-exposed cycle that adds roughly 336 hours and thousands of dollars to every home. Foundation 3D-prints the foundation wall directly from a downloaded build plan, using common, locally sourced concrete instead of expensive proprietary mortar and an external material-flow gate that produces clean starts, stops, and consistent layers. Foundation is an Ohio company commercializing this printer for the U.S. residential-construction market. This TVSF project will validate a production-ready printer and deliver third-party performance data that de-risks a \$23.7B U.S. formwork opportunity while advancing Ohio’s housing-affordability and advanced-manufacturing agendas.

## 1. Market Opportunity

Every residential concrete foundation in the United States is formed with temporary wooden or aluminum formwork that must be built, set, stripped, repaired, and cleaned for reuse. On a typical 2,100 sq ft home this formwork cycle consumes roughly 336 labor hours and a meaningful share of the foundation budget, and it is a chronic source of schedule slip because it cannot proceed in poor weather. The pain is felt hardest by production homebuilders and foundation subcontractors, who face a persistent skilled-labor shortage and thin margins.

Approximately 1.4 million new single-family homes are started in the U.S. each year, essentially all requiring a formed concrete foundation, and the current U.S. spend on residential formwork and foundation labor is estimated at over \$23.7B annually. The addressable pain is the formwork portion of that spend — the build-strip-repair-clean cycle — which is precisely what a direct-printed wall eliminates.

## 2. Overview of Technology/Product

Foundation’s system prints a home’s foundation-wall geometry directly: a build plan is downloaded to the printer, concrete is pumped into the machine, and the nozzle lays concrete one layer at a time while a printing trolley moves laterally and vertically along runway rails. Two differentiators define the product. First, it uses common, locally sourced concrete rather than an expensive proprietary mortar — the single largest cost and supply-chain constraint on competing printers. Second, an external gate controls material flow at the nozzle, enabling clean starts and stops and consistent layers without the smearing and cold-joint defects that plague mortar-extrusion systems. The net effect eliminates the build-strip-repair-clean formwork cycle, cutting foundation labor from ~336 hours to ~25 and formwork/material cost by 47–65% versus both conventional forming and mortar-based concrete printers.

| Performance factor                            | Foundation | Conventional formwork | Mortar-extrusion printers |
|-----------------------------------------------|------------|-----------------------|---------------------------|
| Common local concrete (no proprietary mortar) | ✓          | ✓                     | ✗                         |
| Clean starts/stops (external gate)            | ✓          | n/a                   | ✗                         |
| Eliminates form build-strip-clean cycle       | ✓          | ✗                     | ✓                         |
| Foundation labor hours (2,100 sq ft)          | ~25        | ~336                  | ~120                      |
| Formwork + material cost vs conventional      | –47–65%    | baseline              | –10–20%                   |

## 3. Development Stage and Timeline

The core printing system is beyond proof-of-concept. Foundation has printed full-height wall segments with the runway-rail trolley and the external-gate nozzle using standard ready-mix concrete, validating the layer geometry and clean start/stop behavior that differentiate the product (TRL 5–6). The customer requirement driving the current MVP came directly from production-builder interviews: a printer that runs on locally batched concrete and produces an inspection-ready footing-to-top-of-wall in a single setup. This TVSF project takes the system to a manufacture-ready configuration and a fully instrumented field demonstration.

The 12-month plan below is summarized as a milestone schedule that matches the Q11 milestones one-for-one; time to first commercial installation is estimated at 14 months from award.

**TVSF 12-Month Milestone Schedule (target completion month)**

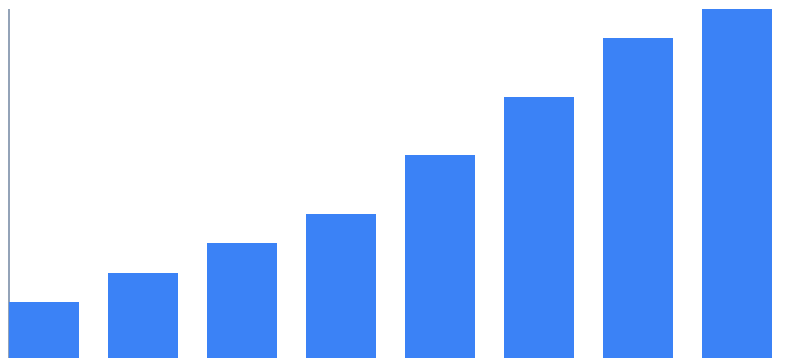


Figure 1. TVSF 12-month milestone schedule — target completion month for MS1–MS8, mapping one-for-one to the Q11 Project Plan table.

## 4. Commercialization and Market Entry Strategy

The residential-construction market adopts new methods cautiously and on evidence, but foundation subcontractors and production builders are actively seeking labor-saving methods because of the skilled-labor shortage. Foundation’s initial target customer is the regional production homebuilder and its foundation subcontractor in high-volume Sunbelt and Midwest markets, where repeatable floor plans make printed foundations especially efficient.

Go-to-market is a printer-plus-service model: Foundation places printers with foundation subcontractors under a lease-plus-per-linear-foot arrangement, with training and remote build-plan support. Foundation has held discovery conversations with regional builders and foundation subs who have confirmed the labor and schedule pain and a willingness to pilot. This answer works in concert with the Q6 business model and the Q7 financial stage; the market-size argument is established in Q1 and is not repeated here.

## 5. IP Position

The technology that is the subject of this TVSF application is protected by a field-of-use exclusive license to a patent owned by [Ohio research institution / federal laboratory], covering the external material-flow gate and layer-control method for cementitious printing (U.S. Patent [number], issued [date]). Foundation has submitted its license application and commercialization plan to the institution’s technology-transfer office, which has assisted in finalizing terms; a definitive field-of-use exclusive agreement for residential foundations is anticipated within [timeframe]. This IP protects the core differentiator — clean, defect-free layers using common concrete — against the mortar-extrusion approaches competitors rely on. In addition, Foundation holds its own provisional filings on the runway-rail trolley kinematics and build-plan pipeline.

## 6. Business Model

Foundation generates revenue two ways: printer leases to foundation subcontractors (recurring monthly), and a per-linear-foot build fee tied to each printed foundation (usage-based). Hardware is deliberately not sold outright — the lease-plus-usage model lowers the customer’s adoption barrier and gives Foundation a recurring, expanding revenue base as each printer prints more homes. R&D remains focused on print speed, concrete-mix latitude, and multi-story capability. The abbreviated pro-forma below uses the required TVSF categories without modification; figures are directional and in \$1,000s.

| (\$1,000s)                | 2026  | 2027    | 2028  | 2029  | 2030   |
|---------------------------|-------|---------|-------|-------|--------|
| Revenues                  |       |         |       |       |        |
| Product sales (printers)  | 0     | 0       | 0     | 0     | 0      |
| Licensing                 | 0     | 0       | 0     | 0     | 0      |
| Printer lease + build fee | 120   | 900     | 3,400 | 9,200 | 21,000 |
| Other                     | 0     | 0       | 0     | 0     | 0      |
| Total revenues            | 120   | 900     | 3,400 | 9,200 | 21,000 |
| Production Expenses       |       |         |       |       |        |
| Cost of Goods Sold        | 90    | 520     | 1,600 | 3,900 | 8,200  |
| Gross profit              | 30    | 380     | 1,800 | 5,300 | 12,800 |
| Other Expenses            |       |         |       |       |        |
| R & D, including IP       | 600   | 900     | 1,300 | 1,800 | 2,400  |
| Sales, General & Admin.   | 350   | 700     | 1,400 | 2,600 | 4,300  |
| Total other expenses      | 950   | 1,600   | 2,700 | 4,400 | 6,700  |
| Net profit                | (920) | (1,220) | (900) | 900   | 6,100  |
| Equity Investment         | 1,500 | 3,000   | 0     | 0     | 0      |

**Pro-Forma Revenue & Gross Profit, 2026–2030 (\$1,000s)**

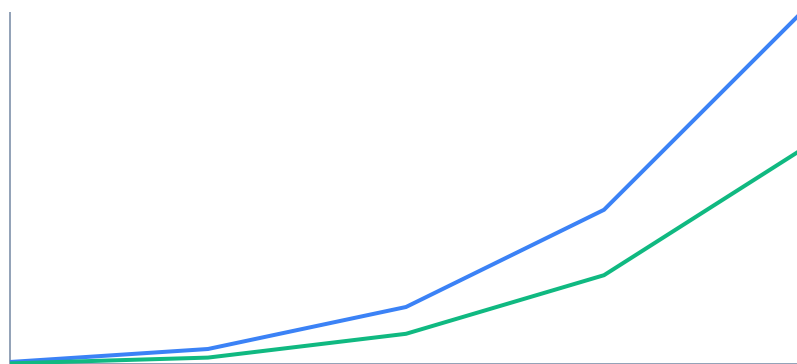


Figure 2. Foundation pro-forma revenue and gross-profit ramp, 2026–2030 (\$1,000s) — the recurring lease-plus-build-fee model visualized from the Q6 pro-forma above.

## 7. Current Financial Stage

Foundation has raised [\$X] to date from [founders / pre-seed angels / an Ohio pre-seed fund] and has [runway]. The company is pre-revenue, with paid pilot commitments contingent on the manufacture-ready printer this TVSF project delivers. The funding strategy is to use the TVSF award and a concurrent [\$1.5M] pre-seed round to reach the first commercial installations, then raise a seed round on the strength of third-party field-performance data. Key assumptions: lease pricing of [\$ / mo], a build fee of [\$ / linear foot], and a printer utilization ramp reflected in the Q6 pro-forma.

## 8. Economic Impact on State of Ohio

Foundation is headquartered in Ohio and will manufacture and assemble its printers in-state, creating skilled advanced-manufacturing and field-service jobs as the fleet scales. Beyond direct employment, printed foundations directly address Ohio’s housing-affordability agenda by cutting foundation cost and schedule for in-state builders, and position Ohio as a hub for construction-technology manufacturing. By 2030 the Q6 model implies a printer fleet that supports [N] Ohio jobs and materially lowers the delivered cost of entry-level housing across the state.

## 9. Management Team

Foundation’s team pairs construction-domain depth with hardware and go-to-market experience:

- CEO — 3D-construction and robotics background; led the printer architecture and the external-gate nozzle from concept to full-height wall prints.
- CTO / Engineering — cementitious materials and motion-control expertise; owns the concrete-mix latitude and runway-rail kinematics.
- Commercial lead — residential-construction and building-products go-to-market; running the builder/subcontractor discovery and pilot pipeline.
- Advisors — an Ohio production homebuilder and a licensed structural engineer, ensuring code-path and inspection readiness.

The team’s combination of a working printer, direct builder relationships, and Ohio manufacturing intent is the foundation of execution risk mitigation — the highest-weighted TVSF review criterion.

## 10. ESP Engagement

Foundation has worked with [EC / Entrepreneurial Center] since [date] as its Entrepreneurial Services Provider. The EC has advised on customer discovery with production builders, the financial model behind the Q6 pro-forma, and the structuring of this TVSF application. Foundation’s primary EC contact is [name]; the engagement is ongoing and the EC’s support letter accompanies this proposal (Volume 3).

## 11. Project Plan

The 12-month TVSF effort takes the printer to a manufacture-ready configuration and delivers an instrumented, inspection-ready field demonstration with third-party performance data. Milestones are measurable and map one-for-one to the Q3 Gantt.

| Milestone                             | Description / Success Criteria                                                               |
|---------------------------------------|----------------------------------------------------------------------------------------------|
| MS1 — Concrete-mix latitude           | Validate printing across 3 locally batched ready-mix designs; layer adhesion meets spec.     |
| MS2 — Manufacture-ready nozzle/gate   | Externally-gated nozzle finalized for production; clean start/stop verified over 200 cycles. |
| MS3 — Trolley/runway production build | Production runway-rail trolley assembled and calibrated to $\pm 3$ mm placement.             |
| MS4 — Build-plan pipeline             | Foundation-plan $\rightarrow$ print-path software validated on 5 production floor plans.     |
| MS5 — Full foundation print (lab)     | Print a complete footing-to-top-of-wall foundation in a single setup.                        |
| MS6 — Third-party structural testing  | Independent lab confirms compressive/flexural performance vs code minimums.                  |
| MS7 — Field demonstration             | Print an inspection-ready foundation at a partner builder site; local inspector sign-off.    |
| MS8 — Performance data package        | Publish labor-hour, cost, and quality data package for commercialization.                    |

## 12. Budget: Table and Narrative

The budget has been vetted with Foundation’s service providers, who have committed to executing on a fixed-price basis to de-risk the TVSF project. Funds are concentrated on manufacture-ready hardware and independent

validation.

| Spend type                                            | OTF (TVSF) funds | Cost share | Total     |
|-------------------------------------------------------|------------------|------------|-----------|
| Personnel                                             | \$95,000         | \$40,000   | \$135,000 |
| Equipment (printer/nozzle/trolley)                    | \$60,000         | \$20,000   | \$80,000  |
| Supplies (concrete, materials)                        | \$18,000         | \$7,000    | \$25,000  |
| Purchased services (structural testing, PCB/embedded) | \$77,000         | \$8,000    | \$85,000  |
| Total                                                 | \$250,000        | \$75,000   | \$325,000 |

\$77,000 in purchased services covers independent structural testing (MS6) and embedded/PCB support to optimize gate control; personnel covers the engineering effort across MS1–MS8.

## 13. Next Steps

Immediately upon award, Foundation will execute the field-of-use license from [institution], locking access to the core IP, and issue fixed-price purchase orders to its service providers for the manufacture-ready build. In parallel it will close the concurrent pre-seed round and confirm the partner-builder field-demonstration site so MS7 can proceed on schedule. On completion, Foundation will convert its discovery-stage builder relationships into paid pilots using the MS8 performance-data package.

## 14. Major Risks and Mitigation

- Concrete-mix variability across regions — mitigated by MS1 (latitude testing across multiple local ready-mix designs) and the external gate's tolerance to mix variation.
- Code/inspection acceptance — mitigated by MS6 third-party structural testing and MS7 inspector sign-off, plus a licensed structural-engineer advisor.
- Adoption inertia among subcontractors — mitigated by the lease-plus-usage model (low barrier) and the MS8 labor/cost data package that quantifies the savings.
- Supply-chain/manufacturing scale — mitigated by fixed-price service-provider commitments and Ohio-based assembly.

## Willingness-to-License Letter

[Institution letterhead] [Date]

To the Ohio Third Frontier / TVSF Review Committee:

[Institution] owns U.S. Patent [number] ("[title]"), covering the external material-flow gate and layer-control method for cementitious printing. [Institution], through its Office of Technology Transfer, confirms that it is willing to license this technology to Foundation on a field-of-use exclusive basis for residential concrete foundations. A license application and commercialization plan have been submitted and are in final negotiation; we anticipate executing a definitive agreement within [timeframe].

We support Foundation's TVSF application to validate and commercialize this technology.

Sincerely, [Name, Title] — [Institution], Office of Technology Transfer

[This is the required Willingness-to-License letter from the IP-owning institution — replace bracketed fields and attach on institution letterhead, ≤ 1 page.]

## ESP Support Letter

[EC / Entrepreneurial Center letterhead] [Date]

To the TVSF Review Committee:

[EC name] has been working with Foundation since [date] as its Entrepreneurial Services Provider. Over this engagement we have advised the company on customer discovery with production homebuilders, the financial

model underlying its pro-forma, and the preparation of this TVSF proposal. Foundation's 3D-printed-foundation technology addresses a real, quantified market need, and the team is well-positioned to execute the proposed validation project.

We endorse this TVSF application and will continue to support the project through award and commercialization.

Sincerely, [Name, Title] — [EC name]

[This is the required ESP support letter from the Entrepreneurial Services Provider — replace bracketed fields and attach on EC letterhead, ≤ 1 page.]